SC12 - Revision H purchasing BOM

140 fitted parts: 129 factory SMT + one user-soldered MCU + 10 user-fitted through-hole parts. Three loose caps extra. Parts estimate: $26.45/board at 10 boards before SMT attrition and spare MCUs; PCB, assembly, freight and tax extra. JLC Standard PCBA candidate: U5 purchased from DigiKey and hand soldered by user; final simulations and factory quote pending. See docs/FACTORY_ORDER.md.

Qty	MPN / supplier part	References	Value / package	Extended USD first tier
1	0451003.MRL / C13982	F1 (Factory SMT)	3A FAST / Fuse_Littelfuse-NANO2-451_453	0.2556
4	0805W8F1001T5E / C17513	R4, R110, R210, R310 (Factory SMT)	1k / R_0805_2012Metric	0.0168
12	0805W8F1002T5E / C17414	R3, R21, R22, R40, R41, R105, R205, R305, R52, R71, R75, R31 (Factory SMT)	10k / R_0805_2012Metric	0.0468
8	0805W8F1003T5E / C149504	R50, R111, R112, R211, R212, R311, R312, R70 (Factory SMT)	100k / R_0805_2012Metric	0.0472
1	0805W8F1152T5E / C17418	R61 (Factory SMT)	11.5k / R_0805_2012Metric	0.0041
3	0805W8F2203T5E / C17556	R101, R201, R301 (Factory SMT)	220k / R_0805_2012Metric	0.015
1	0805W8F3300T5E / C17630	R30 (Factory SMT)	330 / R_0805_2012Metric	0.0054
1	0805W8F5231T5E / C17739	R51 (Factory SMT)	5.23k / R_0805_2012Metric	0.0023
1	0805W8F5902T5E / C17763	R10 (Factory SMT)	59k / R_0805_2012Metric	0.0023
1	1206W4F1200T5E / C17909	R42 (Factory SMT)	120 / R_1206_3216Metric	0.0074
3	1824420000 / 281-2020-ND	J1, J2, J3 (Hand solder)	2P PUSH 3.5mm / Weidmuller_LSF_90_2P	4.89
3	1824430000 / 281-2021-ND	J4, J5, J6 (Hand solder)	3P PUSH 3.5mm / Weidmuller_LSF_90_3P	6.63
1	ABM3B-8.000MHZ-B2-T / C83312	Y1 (Factory SMT)	8MHz 18pF / Abracon_ABM3B_5.0x3.2mm	0.5892
3	AO3401A / C15127	Q101, Q201, Q301 (Factory SMT)	AO3401A / SOT-23	0.2829
1	AO4485 / C51499	Q1 (Factory SMT)	AO4485 / SOIC-8_3.9x4.9mm_P1.27mm	0.2951
1	AP2112K-3.3TRG1 / C51118	U2 (Factory SMT)	AP2112K-3.3 / SOT-23-5	0.1712
2	B240A-13-F / C92506	D4, D5 (Factory SMT)	B240A / D_SMA	0.1902
3	BAT46W-E3-18 / C5557780	D102, D202, D302 (Factory SMT)	BAT46W / D_SOD-123	0.3927
5	BSS138BK,215 / C282529	Q102, Q202, Q302, Q2, Q3 (Factory SMT)	BSS138BK / SOT-23	0.3345
3	CC0805JRNPO9BN102 / C62774	C101, C201, C301 (Factory SMT)	1nF / C_0805_2012Metric	0.0411
2	CC0805JRNPO9BN300 / C107114	C25, C26 (Factory SMT)	30pF / C_0805_2012Metric	0.0146
1	CC0805KRX7R9BB333 / C107150	C69 (Factory SMT)	33nF / C_0805_2012Metric	0.012
2	CL21B103KBANNNC / C1710	C72, C73 (Factory SMT)	10nF / C_0805_2012Metric	0.026
14	CL21B104KCFNNNE / C28233	C3, C6, C11, C20, C21, C22, C27, C40, C41, C50, C51, C66, C74, C75 (Factory SMT)	100nF / C_0805_2012Metric	0.5698

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5	CL21B105KBFNNNE / C28323	C7, C23, C61, C65, C71 (Factory SMT)	1uF / C_0805_2012Metric	0.2165
1	CL21B473KBCNNNC / C53134	C70 (Factory SMT)	47nF / C_0805_2012Metric	0.0173
2	CL21B475KAFNNNE / C98195	C8, C24 (Factory SMT)	4.7uF / C_0805_2012Metric	0.1216
8	CL31A226KAHNNNE / C12891	C4, C5, C13, C12, C60, C62, C63, C64 (Factory SMT)	22uF / C_1206_3216Metric	1.3552
1	CL32B106KBJNNWE / C3844168	C2 (Factory SMT)	10uF / C_1210_3225Metric	0.7145
1	EEEFT1H470AP / C92059	C1 (Factory SMT)	47uF / CP_Panasonic_FT_6.3x5.8	0.3573
1	ESD2CAN24DBZRQ1 / C5736151	D20 (Factory SMT)	ESD2CAN24-Q1 / SOT-23	0.3486
1	LM339LVDR / C22427665	U7 (Factory SMT)	LM339LV / SOIC-14_3.9x8.7mm_P1.27mm	0.5859
1	LMR51420XDDCR / C5246146	U1 (Factory SMT)	LMR51420XDDCR / TSOT-23-6	1.0221
3	M22-2010305 / 952-2274-ND	JP101, JP201, JP301 (Hand solder)	5V / 12V / PinHeader_1x03_P2.00mm_Vertical	0.66
1	MMSZ12T1G / C72599	D3 (Factory SMT)	12V ZENER / D_SOD-123	0.0723
1	PM254V-12-10P-H85 / C492399	J8 (Hand solder)	2x5 SERVICE SOCKET 2.54mm / Service_2x05_P2.54mm	0.1158
1	RC0805FR-0713KL / C137592	R63 (Factory SMT)	13k / R_0805_2012Metric	0.0055
3	RC0805FR-0720KL / C114566	R103, R203, R303 (Factory SMT)	20k / R_0805_2012Metric	0.0237
1	RC0805FR-072K61L / C273965	R62 (Factory SMT)	2.61k / R_0805_2012Metric	0.0075
3	RC0805FR-072M2L / C137552	R104, R204, R304 (Factory SMT)	2.2M / R_0805_2012Metric	0.0219
1	RC0805JR-070RL / C96345	R20 (Factory SMT)	0 / R_0805_2012Metric	0.0066
3	RC1206FR-0710KL / C132649	R109, R209, R309 (Factory SMT)	10k / R_1206_3216Metric	0.0333
6	RT0805BRD07100KL / C122537	R60, R65, R66, R68, R72, R73 (Factory SMT)	100k / R_0805_2012Metric	0.2622
1	RT0805BRD07102KL / C865126	R1 (Factory SMT)	102k / R_0805_2012Metric	0.0471
1	RT0805BRD0712K1L / C865169	R69 (Factory SMT)	12.1k / R_0805_2012Metric	0.0533
1	RT0805BRD0713K7L / C865183	R2 (Factory SMT)	13.7k / R_0805_2012Metric	0.0468
2	RT0805BRD0720KL / C469660	R67, R74 (Factory SMT)	20k / R_0805_2012Metric	0.1486
1	SK6805-EC15 / C2890035	D10 (Factory SMT)	SK6805-EC15 / LED_SK6805_EC15	0.0995
1	SMAJ13A-13-F / C110519	D6 (Factory SMT)	SMAJ13A / D_SMA	0.1241
3	SMAJ30A / C148230	D101, D201, D301 (Factory SMT)	SMAJ30A / D_SMA	0.4362
1	SMBJ22A-E3/52 / C3698580	D2 (Factory SMT)	SMBJ22A / D_SMB	0.4671
1	SN74AHCT1G125DBVR / C7484	U10 (Factory SMT)	SN74AHCT1G125 / SOT-23-5	0.1001
1	SRP5030TA-3R3M / C2042118	L2 (Factory SMT)	3.3uH / L_Bourns_SRP5030T	2.3956

Qty	MPN / supplier part	References	Value / package	Extended USD first tier
1	SRP7028A-100M / C2687402	L1 (Factory SMT)	10uH / L_Bourns_SRP7028A_7.3x6.6mm	0.9358
1	STM32F042K6T6 / 497-14647-ND	U5 (Hand solder)	STM32F042K6T6 / LQFP-32_7x7mm_P0.8mm	4.34
1	TCAN3413DR / C22433320	U6 (Factory SMT)	TCAN3413DR / SOIC-8_3.9x4.9mm_P1.27mm	1.0156
1	TPS2553DBVR-1 / C111738	U4 (Factory SMT)	TPS2553DBVR-1 / SOT-23-6	0.2912
1	TPS259620DDAR / C2155773	U9 (Factory SMT)	TPS259620DDAR / TPS2596_DDA8	0.8154
1	TPS61085DGKR / C113659	U8 (Factory SMT)	TPS61085DGKR / VSSOP-8_3x3mm_P0.65mm	0.6006